

NEXT-GENERATION SEQUENCING SERVICE REPORT

DVPROCAGRF24010052-1

AGRF BIOINFORMATICS

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AGRF NEXT-GENERATION SEQUENCING

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CONTENTS

Bioinformatics Analyst details	1
Project Overview	2
Primary analysis	2
Data Yield	2
Data Delivery	3
AGRF Data Verification	4
Windows:	4
Unix:	4

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Project Overview

Project code: DVPROCAGRF24010052-1
Type of service: Illumina Sequencing
Project start date: Monday 19th February, 2024

This document reports the data generation of the Illumina sequencing run and follow up analysis if applicable.

Primary analysis

Image analysis was performed in real time by the NextSeq Control Software (NCS) v1.5.0.42699 and Real Time Analysis (RTA) v3.10.30, running on the instrument computer. RTA performs real-time base calling on the NextSeq instrument computer. Then the Illumina DRAGEN BCL Convert 07.021.645.4.0.3 pipeline was used to generate the sequence data. The data generated here meet the AGRF quality standards.

Data Yield

The data yield is as follows:

Table 1:

300bp Paired End - Flowcell ID: AACLLM5M5

Lane	Sample Name	Paired End	Data Yield(bp)
1	C13-16S_V3-V4	653,923	0.39 Gb
	C14-16S_V3-V4	276,926	0.17 Gb
	C15-16S_V3-V4	31,784	0.02 Gb
	M13-16S_V3-V4	318,220	0.19 Gb
	M13C-16S_V3-V4	812,040	0.49 Gb
	M13N-16S_V3-V4	723,864	0.44 Gb
	M13P-16S_V3-V4	330,990	0.20 Gb
	M13PN-16S_V3-V4	498,644	0.30 Gb
	M14-16S_V3-V4	214,513	0.13 Gb
	M15-16S_V3-V4	408,256	0.25 Gb

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Table 1 – Continued from previous page

Lane	Sample Name	Paired End	Data Yield(bp)
	U13-16S_V3-V4	419,706	0.25 Gb
	U13C-16S_V3-V4	540,904	0.33 Gb
	U13N-16S_V3-V4	239,162	0.14 Gb
	U13P-16S_V3-V4	270,011	0.16 Gb
	U13PN-16S_V3-V4	361,967	0.22 Gb
	U13f-16S_V3-V4	549,471	0.33 Gb
	U14-16S_V3-V4	231,280	0.14 Gb
	U14f-16S_V3-V4	505,900	0.30 Gb
	U15-16S_V3-V4	333,321	0.20 Gb
	U15f-16S_V3-V4	581,774	0.35 Gb
	blank-16S_V3-V4	188,779	0.11 Gb
Total		8,491,435	5.11Gb

Data Delivery

The data has been transferred and made available to you.

Below is a brief description of the files to expect.

<sample_name>_<flowcell_ID>_<index>_<lane>_<readNum>_fastq.gz - compressed FastQ formatted sequence file containing untrimmed 300 bp reads, where <readNum> specifies the first or second read of the pair.

These files contain the read **sequence output** with the corresponding Illumina **quality scores**. The quality is encoded in symbolic ASCII format in the following way: char (phred-like quality score + 33).

The files have been compressed using gzip and can be uncompressed using gunzip (*nix) or 7-Zip (Windows).

Please review the sequencing files after you have received them. Files will **remain on the server** until the **20th of March, 2024** . After that date they will be removed from the server.

A backup copy will be archived at AGRF until the **19th of May, 2024** . Please note that charges will apply for restoring files to the server.

If you have any questions or require further assistance please do not hesitate to con-

tact us at NextGen@agrif.org.au.

AGRF Data Verification

As part of the general quality assurance AGRF tests the integrity and completeness of the data before it is sent to you.

Your packaged data has checksum information provided by AGRF should you wish to verify that your data is complete and error-free. Verification methods for both Windows and Unix systems are outlined below:

Windows:

The checksum information of the files included has been generated using Exactfile, which is a free third-party tool for Windows:
<http://www.exactfile.com/>.

Included in your directory are two files, “checksums.exf”, which contains the MD5 hashes of the original files on AGRF’s data server and “TestFiles.exe”, an applet created by the Exactfile program to check files against the hashes listed in “checksums.exf”.

To test the integrity of your files, you can simply start “TestFiles.exe”.

Please be patient as it may take up to several minutes before the file check report appears, because of the size of the fastq files.

Unix:

The checksum information of the files included has been generated using the md5sum program. The calculated checksums of your data files are included in the “checksums.md5” file.

To test the integrity of your files, type the following command while in the directory containing checksums.md5:

```
md5sum -c checksums.md5
```



*Please note: The checksum verification can only be performed on the original data as received from AGRF, before any files are moved or modified. AGRF accepts no responsibility regarding the performance of Exactfile program.